

Problem 1. Does the following sum converge or diverge:

$$\sum_{i=1}^{\infty} \frac{1}{2025^{\ln(n)}}.$$

Problem 2. Does the following sum converge or diverge:

$$\sum_{i=1}^{\infty} \frac{(n+1)!}{10^n}.$$

Problem 3. Let (a_n) be a sequence of positive numbers such that $\sum a_n$ converges. Given an arbitrary sequence, (b_n) , of 1's and 0's, does $\sum a_n b_n$ converge?

Problem 4. Does the following series converge or diverge:

$$\sum_{i=1}^{\infty} \frac{1}{2^{n-(-1)^n}}.$$

Problem 5. Does the following series converge or diverge:

$$\sum_{i=1}^{\infty} \frac{2^n + \cos(\pi n)\sqrt{n}}{3^{n+1}}.$$